Quantification: 682_z29
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	12
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1740.55	1624	939.642	px^2
major_axis_length_px	lacuna shape	75.1181	75.9278	23.0542	px
minor_axis_length_px	lacuna shape	29.0371	30.0551	9.2359	px
aspect_ratio	lacuna shape	2.7064	2.5263	0.7296	unitless
eccentricity	lacuna shape	0.9096	0.9183	0.0622	unitless
solidity	lacuna shape	0.87	0.8778	0.0425	unitless
roots_count	headline	5.3333	5	2.9336	count
ring_length_r30_px	headline	271.333	279	112.766	px
ring_length_r60_px	reported	719.5	745	289.725	px
owned_length_px	ownership-dependent	1833.15	1089.91	1865.4	px
edge_count	ownership-dependent	80.25	39.5	86.6898	count
mean_edge_length_px	ownership-dependent	27.9834	24.0785	9.0607	px
perimeter_px	normalised	188.488	208.43	72.9325	px
ring_area_r30_px2	normalised	7625.92	8153	2258.11	px^2
ring_area_r60_px2	normalised	19043.3	19608.5	5099.65	px^2
in_frame_fraction_r30	normalised	0.9948	1	0.018	unitless
in_frame_fraction_r60	normalised	0.9826	1	0.0559	unitless
ring_density_r30	normalised	0.035673	0.0346775	0.0108953	px^-1
ring_density_r60	normalised	0.0376223	0.0383353	0.00985305	px^-1
roots_per_100px_perimeter	normalised	2.8718	2.437	1.3562	per 100 px
ring_attached_length_r30_px	network v2	202.167	175.5	115.365	px
ring_attached_length_r60_px	network v2	470.083	338.5	350.867	px
ring_length_w_r30_px	network v2	302.026	307.101	123.452	px
ring_length_w_r60_px	network v2	805.309	825.991	323.823	px
sholl_crossings_r10	network v2	6	6	2.8284	count
sholl_crossings_r20	network v2	9.3333	10.5	3.8218	count
sholl_crossings_r30	network v2	11.1667	10.5	3.9505	count
ring_width_mean_r30_px	width	6.9667	7.0528	0.8127	px

Per field
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measure	value	units
analysed_area_px2	1.02339e+06	px^2
total_skeleton_length_px	33508	px
canalicular_length_density_per_px	0.0327421	px^-1
junction_count	674	count
junction_density_per_px2	0.00065859	px^-2
lacuna_count	14	count
lacunae_per_px2	1.368e-05	px^-2
skeleton_component_count	265	count

mean_component_length_px	126.445	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0370076	px^-1
field_density_without_flagged_per_px	0.0330575	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	513	px
unattached_component_count	32	count
canalicular_length_density_attached_per_px	0.0327421	px^-1
field_length_density_with_unattached_per_px	0.0332434	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4342	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	31329	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	446	49.6136	12.2658	4.0449	0.969	0.7866	-1.4445	249.987	449.359	False
4	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
5	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
6	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
7	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
8	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
9	513	41.8409	17.3844	2.4068	0.9096	0.8607	1.3405	830.565	75.5867	False
10	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
11	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
12	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
13	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False
14								548.08	212.663	False

Network per lacuna

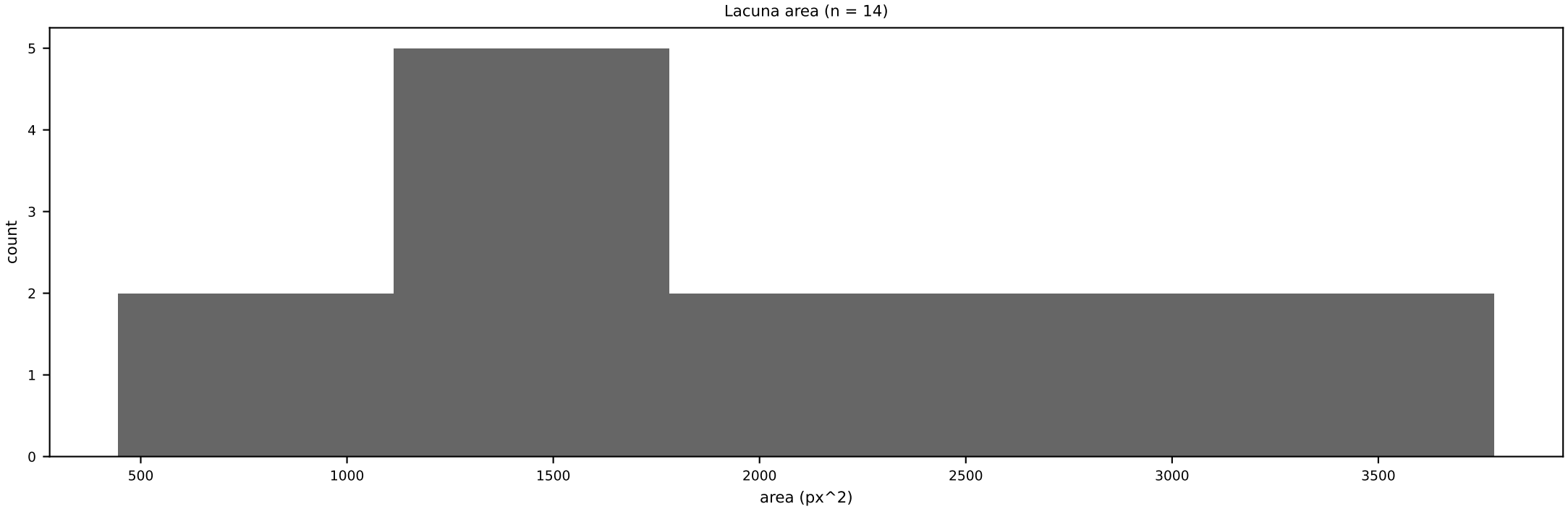
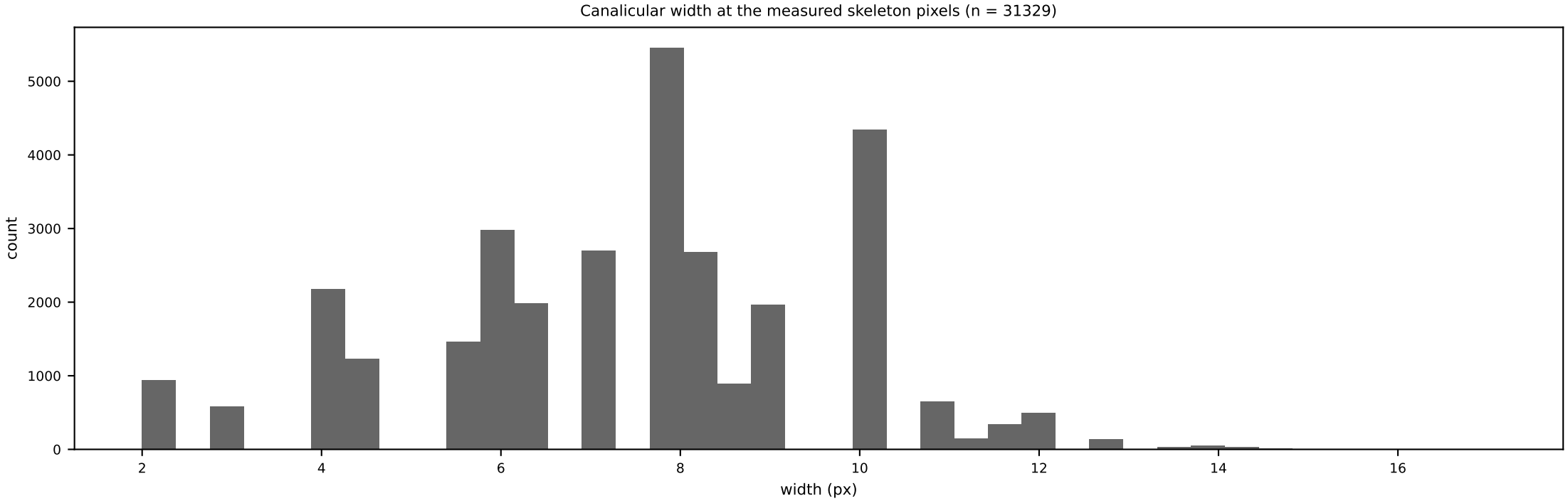
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	128	230	90.8406	8	11.3551	
2	10	463	1191	4157.8	190	21.8831	7.0028
3	5	347	918	5123.55	233	21.9895	7.4637
4	6	393	1192	4365.25	206	21.1905	7.5622
5	5	268	796	2413.55	100	24.1355	6.8365
6	7	394	858	2668.29	115	23.2025	7.3633
7	9	361	783	1341.7	55	24.3946	7.473
8	4	298	758	1507.23	51	29.5535	5.2862
9	5	155	446	314.451	11	28.5864	6.8163
10	3	155	439	340.149	10	34.0149	7.9932

11	1	118	389	52.3553	1	52.3553	5.5177
12	7	290	732	672.6	28	24.0214	7.0222
13	10	218	381	348.22	17	20.4835	7.0835
14	1	157	534	34.3848	1	34.3848	7.6531

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0228286	0.0166437	1.1446
2	229.522	9151	23871	1	0.999	0.0505956	0.0498932	4.3569
3	120.811	6134	17911	1	1	0.0565699	0.0512534	4.1387
4	309.279	11331	27269	1	1	0.0346836	0.0437126	1.94
5	235.238	9387	21888	1	1	0.0285501	0.036367	2.1255
6	214.167	8637	19737	1	1	0.0456177	0.0434717	3.2685
7	322.836	9940	21386	0.8402	0.7724	0.0363179	0.0366127	2.7878
8	229.48	8595	22042	1	1	0.0346713	0.0343889	1.7431
9	109.498	4245	11066	1	0.9865	0.0365135	0.0403036	4.5663
10	161.817	5491	13376	1	1	0.028228	0.03282	1.8539
11	147.196	6914	19480	1	1	0.0170668	0.0199692	0.6794
12	254.693	9728	23664	1	1	0.0298109	0.0309331	2.7484
13	202.693	7711	16371	0.9376	0.8055	0.0282713	0.0232729	4.9336
14	47.4558	4187	11845	1	1	0.037497	0.0450823	2.1072

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	20	20	145.397	253.652	1	4	4
2	377	1057	509.534	1347.5	8	16	18
3	308	902	390.078	1018.34	8	11	11
4	353	1007	437.635	1335.45	11	13	16
5	167	356	292.096	878.742	5	12	12
6	317	662	437.978	959.968	8	11	15
7	350	686	404.492	880.997	10	12	11
8	170	261	328.208	836.127	8	12	10
9	140	321	173.024	496.333	6	4	5
10	133	299	179.681	491.09	4	7	8
11	46	96	136.196	452.073	2	5	8
12	213	409	322.108	815.855	5	10	14
13	181	241	242.894	425.404	6	6	7
14	21	30	174.882	606.831	1	5	10

682_z29: distributions, pre-validation, px